



G133HAN01.1

EC Notice

AUO Display Plus

2024/8/14

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ECN for LED Driver IC & Connector



- **Purpose :** Since the current LED driver IC will be phase out, it will be replaced by a new IC. In addition, there is an error in the connector type information on the product specification, so we will correct it.
- **Change Items:**

Items	Before	After
LED driver IC	RT8577AGQW, Richtek	NT50508VG/B , Novatek
Connector_1	DF19K-30P-1H (54)	DF19K-30P-1H (52)
Connector_1 Mating Housing	DF19G-30S-1C(05) DF19A-2830SCFA(41)	DF19G-30S-1C(06) DF19A-2830SCFA(41)
Connector_2	DF19G-8P-1H(54)	DF19G-8P-1H(52)
Connector_2 Mating Housing	DF19G-8S-1C(05) DF19A-2830SCFA(41)	DF19G-8S-1C(06) DF19A-2830SCFA(41)

- **Implement data and method:** Running change from 2024/I2

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- **Spec:** The LED driver FPWM Dimming Duty cycle will be changed as below table.

Before

Parameter	Symbol	Min	Typ	Max	Units	Remark
LED Power Supply	VLED	10.8	12.0	13.2	[Volt]	Define as Connector Interface (Ta=25°C)
LED Input current	ILED	-	-	1.25- (Note 1)	[A]	
LED Enable Input High Level	LED_EN	2	-	5	[Volt]	
LED Enable Input Low Level		-	-	0.5	[Volt]	
PWM Logic Input High Level	LED_PWM	2	-	5	[Volt]	
PWM Logic Input Low Level		-	-	0.5	[Volt]	
PWM Input Frequency	FPWM	500	-	10K	Hz	
PWM Duty Ratio	Duty	5	-	100	%	
LED Inrush Current	ILED _{Rush}	-	-	2	[A]	

After

Parameter	Symbol	Min	Typ	Max	Units	Remark
LED Power Supply	VLED	10.8	12.0	13.2	[Volt]	Define as Connector Interface (Ta=25°C)
LED Input current	ILED	-	-	1.25- (Note 1)	[A]	
LED Enable Input High Level	LED_EN	2	-	5	[Volt]	
LED Enable Input Low Level		-	-	0.5	[Volt]	
PWM Logic Input High Level	LED_PWM	2	-	5	[Volt]	
PWM Logic Input Low Level		-	-	0.5	[Volt]	
PWM Input Frequency	FPWM	200	-	18K	Hz	
PWM Duty Ratio	Duty (FPWM 200Hz~5KHz)	5	-	100	%	
	Duty (FPWM 5KHz~18KHz)	15		100	%	
LED Inrush Current	ILED _{Rush}	-	-	2	[A]	

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- **Spec:** The LCM connector will be changed as below table.

Before

7.1.1 Connector(CN1)

Connector Name / Designation	Interface Connector / Interface card
Manufacturer	Hirose
Type Part Number	DF19K-30P-1H (54)
Mating Housing Part Number	DF19G-30S-1C(05) DF19A-2830SCFA(41)

7.2.1 Connector (CN2)

Connector Name / Designation	LED Connector
Manufacturer	Hirose
Type Part Number	DF19G-8P-1H(54)
Mating Housing Part Number	DF19G-8S-1C(05) DF19A-2830SCFA(41)

After

7.1.1 Connector(CN1)

Connector Name / Designation	Interface Connector / Interface card
Manufacturer	Hirose
Type Part Number	DF19K-30P-1H (52)
Mating Housing Part Number	DF19G-30S-1C(06) DF19A-2830SCFA(41)

7.2.1 Connector (CN2)

Connector Name / Designation	LED Connector
Manufacturer	Hirose
Type Part Number	DF19G-8P-1H(52)
Mating Housing Part Number	DF19G-8S-1C(06) DF19A-2830SCFA(41)

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➤ RA test Result

ITEMS	Required Condition	Result
Optical Measurement	Brightness , Chromaticity , Contrast, Uniformity.....	Pass
Temperature Humidity Bias	Ta= 40°C, 90%RH, 300hours	Pass
Low Temperature Operation	Ta= 0°C, 300hours	Pass
Thermal Shock Test	-20°C/30min, 60°C/30min, 100 cycles	Pass
ESD	Contact Discharge: $\pm 8\text{KV}$, 150pF(330 Ω) 1sec, 8 points, 25 times/ point. Air Discharge: $\pm 15\text{KV}$, 150pF(330 Ω) 1sec 8 points, 25 times/ point.	Class B Pass
EMI	Limit/CISPR32 CLASS-B.csv 10m factor/T65(60)2017.csv horizontal Limit/CISPR32 CLASS-B.csv 10m factor/T65(60)2017.csv vertical	Pass

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